

Simple SECR workflow

Wrap-up session

Centre for Research into Ecological & Environmental Modelling
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Wrap-up

A standardized workflow for designing and analyzing camera trap surveys of large cats with spatially explicit capture-recapture (SCR).

- Session 1: Concepts
- Session 2: Simple workflow
- Session 3: Adding complexity
- Session 4: Automatic checks of data integrity
- Session 5: Wrap up

SCR with count traps: Individuals $\times$ Traps $\times$ Sessions

Occasion 1																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
IND1	3	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0
IND2	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0
IND3	0	0	0	0	0	0	0	2	0	0	0	2	0	0	0	0
IND4	0	0	0	0	1	4	0	0	0	0	0	0	0	0	0	0
IND5	0	0	0	0	0	0	0	0	0	5	2	0	0	1	0	0
IND6	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0
IND7	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0
IND8	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

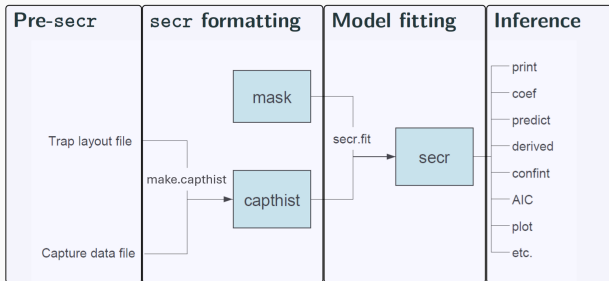
Occasion 1																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
IND1	2	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0
IND2	0	0	0	3	1	0	0	0	0	0	0	0	0	0	0	0
IND3	0	0	0	0	0	0	0	3	0	0	0	1	0	0	0	0
IND4	0	0	0	0	1	4	0	0	0	0	0	0	0	0	0	0
IND5	0	0	0	0	0	0	0	0	0	0	4	1	0	0	1	0
IND6	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0
IND7	0	0	0	0	0	0	0	0	0	0	0	0	3	2	0	0
IND8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0

Session 1

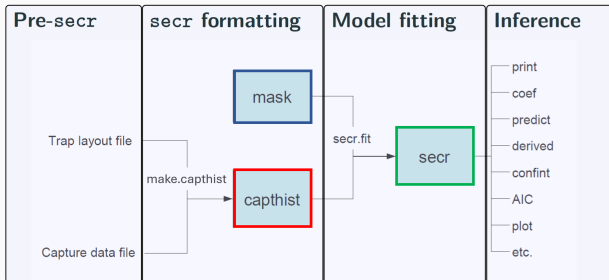


Session 2

secr basic workflow



secre Adding complexity



1. Longer survey periods and closure
2. Shortened-survey and multisession alternatives
3. Elevation constraints and reporting masks
4. Gorge-avoiding movement distances
5. Sex-specific model components

secre Data checking

SCR Data Checker

Column Names

Data Checks

Report

Both files are approved. Detailed checks and report download are available below.

Download HTML report

Traps File

Errors: 2 | Warnings: 2

session

trapID

Coordinates

Effort

Dates

Covariates

- Warning trapID contains whitespace or punctuation that may not parse reliably in secre. Problem values: Camera&11, TPC 38
- Warning Possible swapped coordinates detected for: Camera8
- Error start_date could not be parsed for all rows. Some values do not match the inferred date format. 1312/2022, 2023/29/05
- Error end_date could not be parsed for all rows. Some values do not match the inferred date format. 2023/29/06

Data type: numeric

Detections File

Errors: 0 | Warnings: 1

session

animalID

occasion

trapID

Date

Covariates

- Warning Sex contains values other than Male/Female or M/F. Problem values: Unknown

Future plans

- Code and vignettes will be maintained on GitHub
- Refined through use in case studies (if interested in collaborating, please get in touch)
- Accompanying paper describing the workflow to be posted on bioArXiv